

July 2, 2026

To whom it may concern,

I'm excited to apply for the Software Engineer position on the Query Integration team at MongoDB. Building query engines is what I do every day, and I'd bring a track record of shipping query engine features that hold up under real enterprise load.

At RelationalAI, I design and implement extensions to our Datalog-based relational query language. Working across the “full stack” of a query, I shape the language around customer needs, write the optimization passes that make it fast, and build the tooling and observability to debug all of it. Recent work includes a GQL-style graph pattern matching extension and a procedural extension that has driven previously intractable queries down to sub-hour runtimes. I also maintain the engine's Protocol Buffer entry point, which serves around 100k queries a day, so I know what it takes to keep a query interface reliable at scale.

This engineering is grounded in programming languages research. I designed and published a probabilistic programming language for optimization under uncertainty that outperforms the state of the art by up to 10x, work that earned first place at the ACM Student Research Competition at a top programming languages conference. At MongoDB, I'm looking forward to taking a query language from theory to implementation at a larger scale than I've worked at before, and I'd love to get to know the team.

Thank you for your consideration,
Minsung Cho